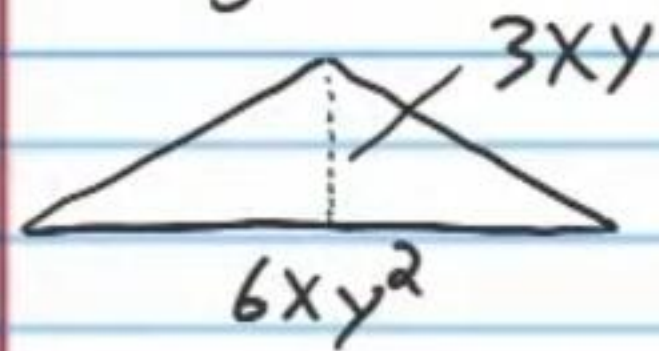


Pre-Algebra Test IV

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show all work.
- iii) Draw a rectangle around your answers.
- iv) Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

- ① Find the area of the triangle below.



- ② If the area of a parallelogram is 6.9 cm^2 and its base is 3 cm , what is its height?

- ③ Find the surface area of a rectangular prism with length 3 meters, height 1 meter, and width 5 meters.

- ④ What percent of 20 is 35?

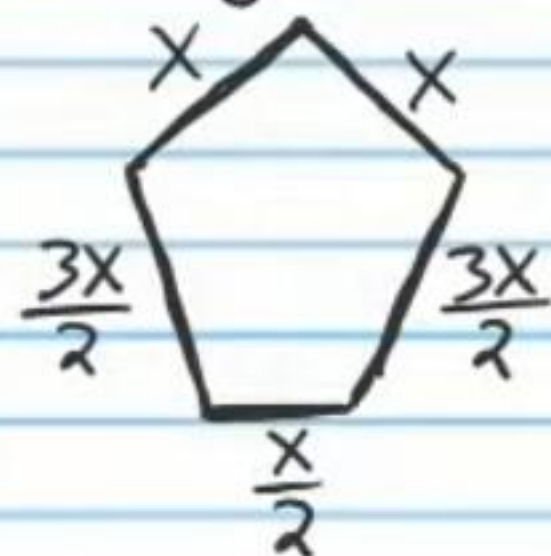
- ⑤ A bank loans you \$3,050 and charges 12% simple interest monthly. Over time, your debt grows to \$6,710. Assuming you didn't return any of the money, how much time passed since you took out the loan?

- ⑥ The sum of five consecutive odd integers is 95. Identify these integers.

- ⑦ Find the average of the following numbers:
 $7x, 3x, 5x, 6x, 3x$.

⑧ If 123 is decreased to 48, what is the percent decrease? Round to the nearest hundredth of a percent.

⑨ Find the perimeter of the figure below.



Graph the following equations. Show all work and identify the points you use to graph the equations. Label your graphs properly.

⑩ $y - x = 3$

⑪ $-y + x = -200$

⑫ $x = -1 + \frac{y}{-4}$

⑬ 9% of what number is 80?

Find the algebraic solutions and the graphical solutions of the following inequalities.

⑭ $5 - 4x > 2(2 + 2x) - 1$ ⑮ $15y - 2 \leq 10y$

⑯ Billy's height grew 3.1%. His current height is 42 inches. What was his height before this increase? Round to the nearest thousandth.

Solve

⑰ $3.9z - 5z + 1.8 = 10.9z$ ⑱ $-5x + 8.2x = 1 + (-4)$

⑲ $-4.8y + 2 = -3.5y - 6.7y$ ⑳ $16.9 - 2(4.3x + 1) = -(2.3x + 4)$

- ②1 a) Evaluate your answer for problem #9 if $x=6$.
b) Evaluate your answer for problem #1 if $x=-1$ and $y=2$.

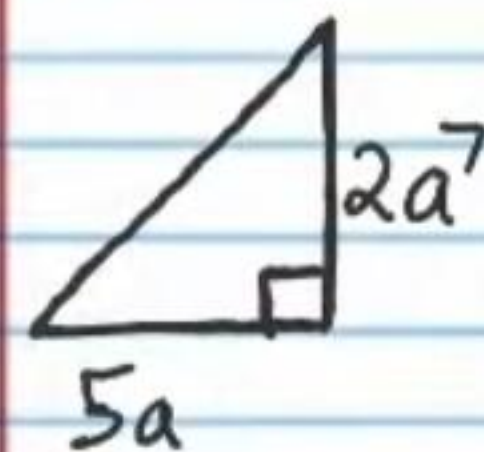
②2 Find the volume of a cylinder with radius $\frac{2x}{5y^4}$ and height $\frac{3x^2}{4y}$. Leave π as a factor.

Graph the following inequalities. Show all work and identify the points you use to graph the inequalities. Label your graphs properly.

②3 $y < -3x$

②4 $\frac{x}{2} - 3y \geq 1$

②5 Find the area of the triangle below.



Pre-Algebra Test IV Answers

① $9x^2y^3$

② 2.3 cm

③ 46 m^2

④ 175%

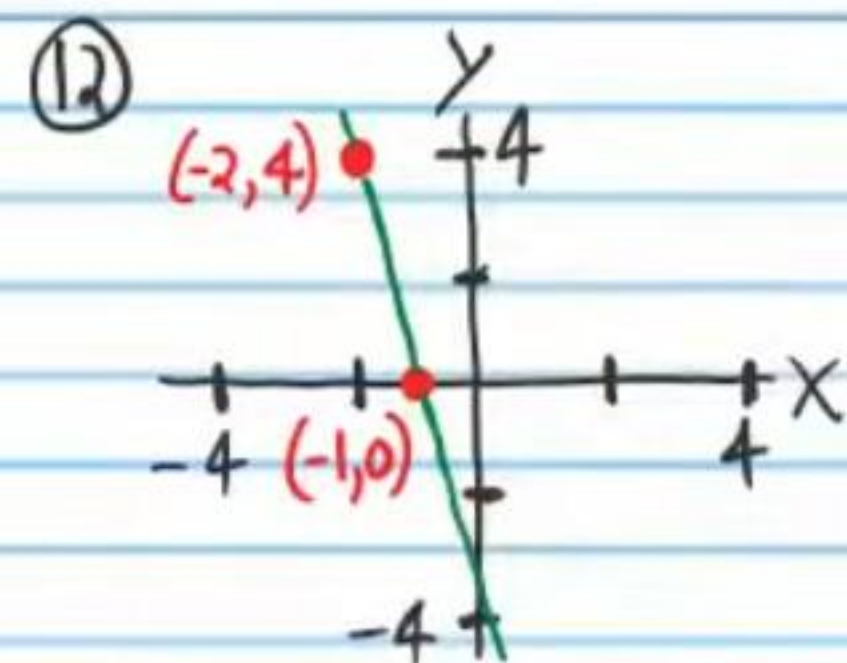
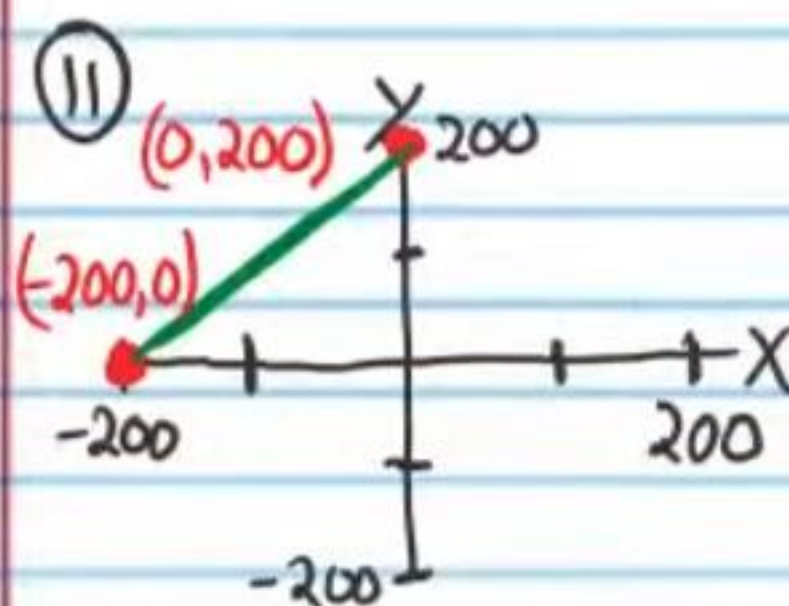
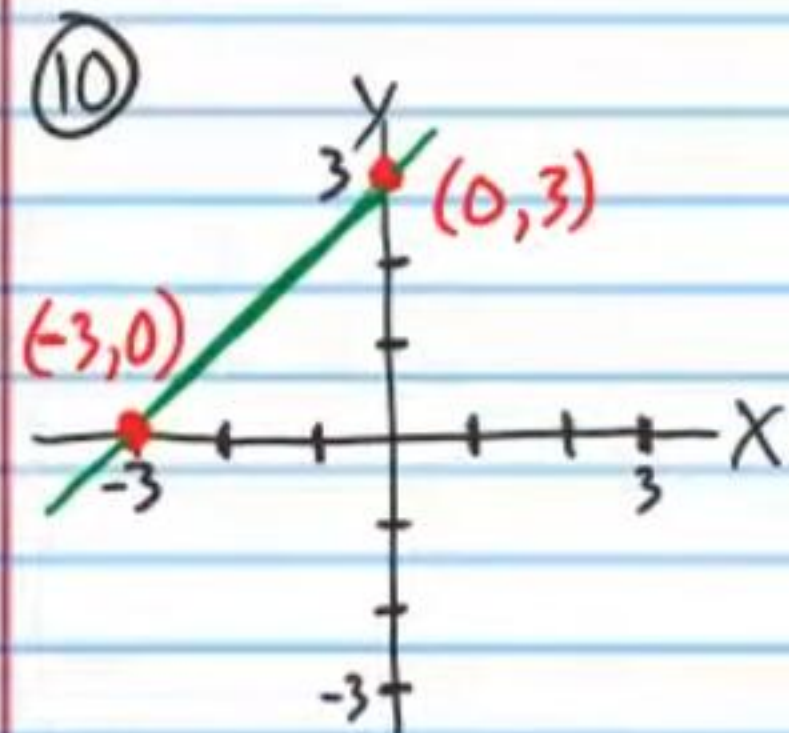
⑤ 10 months

⑥ $15, 17, 19, 21, 23$

⑦ $24x/5$

⑧ 60.98%

⑨ $11x/2$



⑬ $888.\overline{8}$

⑭ $x < \frac{1}{4}$

A number line with tick marks at $-\frac{1}{2}, -\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2}$. An open circle is drawn at $\frac{1}{4}$, and the line is shaded to the left of this point.

⑮ $y \leq \frac{2}{5}$

A number line with tick marks at $-\frac{3}{5}, -\frac{2}{5}, -\frac{1}{5}, 0, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}$. A closed circle is drawn at $\frac{2}{5}$, and the line is shaded to the left of this point.

⑯ 40.737 in

⑰ $z = 0.15$

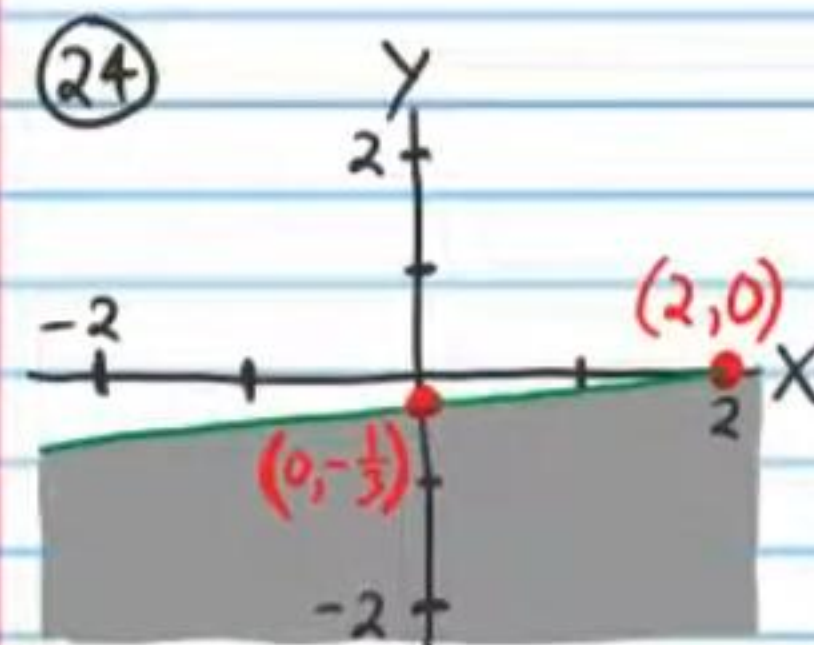
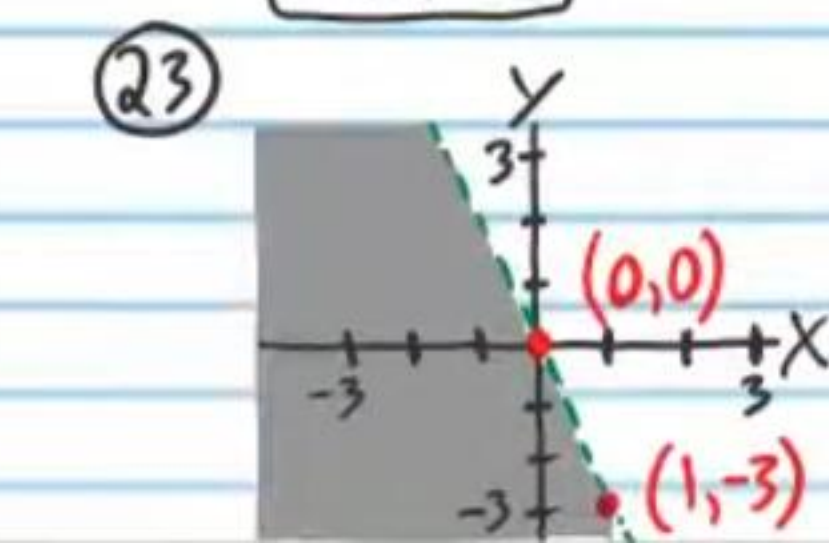
⑱ $x = -0.9375$

⑲ $y = -0.370$

⑳ $x = 3$

㉑ a) 33 b) 72

㉒ $\frac{3\pi x^4}{25y^9}$



㉕ $5a^8$